NEURALPATH CHEATSHEET

Module 7: Reinforcement Learning

Study MDPs, Bellman equations, Q-learning, policy gradients, PPO, bandits, and post-training al

Core Formulas, Mental Models & Engineering Rules

- MDP: (S, A, P, R, gamma). Bellman optimality connects V*(s) and Q*(s,a).
- Q-Learning: Q(s,a) <- Q(s,a) + alpha [r + gamma max Q(s', a') - Q(s,a)].
- PPO clips policy probability ratios within [1-eps, 1+eps] for stable updates.
- Post-training Alignment: DPO optimizes preference pairs directly; GRPO drops the critic network.

Estimated study time: 5 hours · Badge: Policy Optimizer